UNIFIED MULTI-TASK DEEP LEARNING FRAMEWORK FOR PRECISION CATTLE MONITORING

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1. ABSTRACT

Precision livestock farming increasingly relies on computer vision to monitor cattle health, welfare, and productivity. Existing unified cattle-monitoring systems chain task-specific sub-networks without a shared representation, which inflates computational cost and limits real-time on-farm deployment. This synopsis proposes a unified multi-task deep learning framework that jointly learns cattle detection, individual re-identification, and behaviour recognition over a shared backbone, with tracking performed as a downstream association step that reuses the learned embeddings. The framework will be trained with a masked-label heterogeneous multi-task scheme on public datasets and evaluated using mAP, IDF1, HOTA, and behaviour-classification accuracy. The expected outcome is a lightweight, jointly optimised system suitable for continuous on-farm monitoring.

2. INTRODUCTION

Continuous monitoring of individual cattle supports early detection of lameness, oestrus, and abnormal behaviour, directly improving animal welfare and farm economics. Manual observation does not scale to commercial herds, motivating automated vision-based systems.

Most published systems address detection, re-identification, tracking, and behaviour recognition as separate models chained in a pipeline. Each stage recomputes visual features, multiplying inference cost and propagating errors between stages. A shared-representation multi-task design can remove this redundancy.

2.1 Background of the Study

Deep convolutional and transformer-based detectors now achieve high accuracy on livestock imagery, while metric-learning methods enable individual identification from coat patterns. Joint detection-and-embedding models such as FairMOT demonstrate that detection and re-identification can share a backbone, although task competition must be managed.

2.1.1 Scope of the Work

This research is limited to vision-based monitoring of housed dairy and beef cattle using fixed barn cameras and publicly available datasets.

2.1.1.1 Assumptions

Cameras are assumed to be calibrated, fixed, and to provide continuous daytime footage of the monitored pens.

3. LITERATURE REVIEW

Early cattle-monitoring work applied YOLO-family detectors for counting and localisation, followed by appearance-based re-identification using triplet loss on coat patterns. Recent multi-camera studies extend identification across views using self-supervision. Behaviour recognition has progressed from frame-level CNN classification to spatio-temporal models. However, published "unified" systems remain modular pipelines of independently trained sub-networks; no system jointly optimises detection, re-identification, and behaviour over a shared representation while remaining lightweight enough for on-farm deployment.

4. OBJECTIVES OF RESEARCH

The objectives of this research are as follows.

1. To design a shared-backbone multi-task architecture for cattle detection, individual re-identification, and behaviour recognition.
2. To develop a masked-label heterogeneous training scheme enabling joint optimisation across datasets with incomplete label sets.
3. To integrate tracking as a downstream association step that reuses the learned re-identification embeddings.
4. To evaluate the framework against single-task and pipeline baselines on public cattle datasets.

5. RESEARCH QUESTION / PROBLEM

Can a single jointly optimised shared representation support cattle detection, individual re-identification, and behaviour recognition with accuracy comparable to task-specific pipelines, while reducing computational cost sufficiently for real-time on-farm deployment? Existing systems answer monitoring sub-tasks in isolation; the absence of a shared-representation framework constitutes the research gap addressed here.

6. METHODOLOGY (SAMPLING/MEASUREMENT/RESEARCH DESIGN/DATA ANALYSIS)

The proposed framework comprises a shared convolutional backbone with task-specific necks and three heads: a detection head, an ArcFace-based re-identification head, and a spatio-temporal behaviour head. Tracking is performed downstream using BoT-SORT-style association over the learned embeddings. The overall architecture is shown in Figure-6.1.

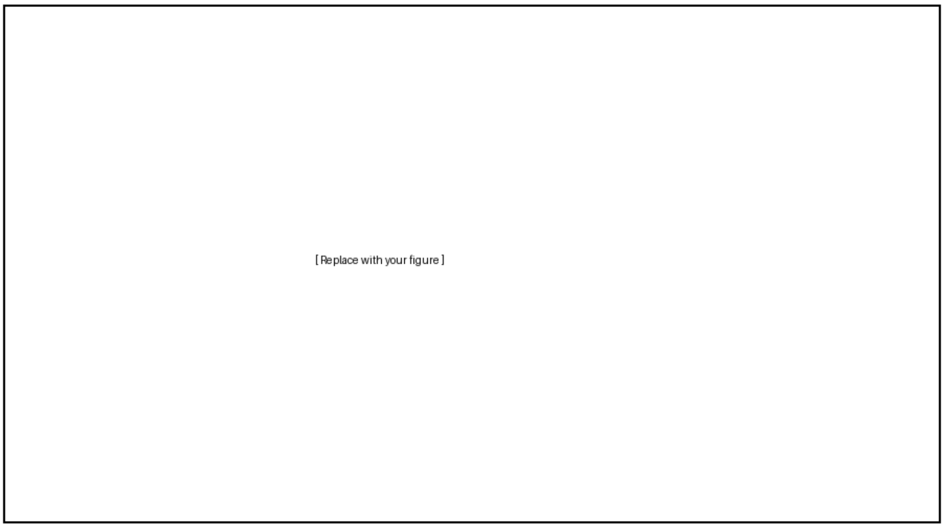

Figure-6.1: Architecture of the proposed unified multi-task framework (Reference if any)

6.1 Datasets

Four public datasets will be used: MultiCamCows2024, Cows2021, CattleBehaviours6, and CVB. Since no single dataset provides all label types per frame, a masked-label training scheme activates only the losses for which ground truth exists in each batch.

6.2 Evaluation Metrics

Detection will be evaluated with mAP; re-identification with rank-1 accuracy and mAP; tracking with HOTA, IDF1, MOTA, and identity switches; behaviour recognition with top-1 accuracy and macro-F1. The plan is summarised in Table-6.1.

Table-6.1: Evaluation metrics and datasets planned for each task of the proposed framework (Reference if any)

Task	Metrics	Dataset
Detection	mAP@0.5, mAP@0.5:0.95	Cows2021
Re-identification	Rank-1, mAP	MultiCamCows2024
Tracking	HOTA, IDF1, MOTA, IDS	Cows2021
Behaviour recognition	Top-1 accuracy, macro-F1	CattleBehaviours6, CVB

6.3 Multi-Task Loss

The framework is trained by minimising a weighted sum of the task losses, where the weights balance gradient magnitudes to mitigate negative transfer:

$$L_{total} = \lambda_{det} L_{det} + \lambda_{id} L_{arc} + \lambda_{beh} L_{beh} \tag{6.1}$$

7. EXPECTED OUTCOMES OF RESEARCH

The research is expected to deliver a jointly optimised, lightweight multi-task framework achieving accuracy comparable to task-specific pipelines at a fraction of the inference cost; empirical evidence on shared versus separate backbones (negative-transfer analysis); a masked-label training recipe for heterogeneous livestock datasets; and a deployable prototype validated on consumer-grade GPU hardware.

8. REFERENCES

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9. APPENDICES

APPENDIX-01

Print of system generated Turnitin similarity report (only ‘originality report’ showing ‘primary sources’) **signed by the author and the supervisor** is required to be placed here.

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